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POWER TOOL

Abstract

A power tool includes a housing, a motor, a fan and a cyclone separation unit. The fan generates an air flow that flows in from first air inlets of the housing and passes through the housing. The cyclone separation unit has second air inlets arranged in a circumferential direction of the cyclone separation unit. The first and second air inlets are arranged apart from each other in an axial direction of the cyclone separation unit. A first dust collecting chamber is defined: (i) between the first air inlets and the second air inlets in the axial direction of the cyclone separation unit, and (ii) between an inner peripheral surface of the housing and the cyclone separation unit, and configured to collect dust by inertia force. A second dust collecting chamber is defined within the cyclone separation unit and configured to collect dust by centrifugal force.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to Japanese patent application No. 2024-030666 and 2024-030667, both of which were filed on Feb. 29, 2024. The contents of the foregoing applications are hereby fully incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a power tool having a cyclone separation unit.

BACKGROUND

[0003] Generally, in processing operation of processing concrete, wood, metal or other materials by using a power tool, a large amount of dust is generated. The dust entering a housing of the power tool may cause a failure or trouble of a mechanism inside the housing. To cope with this problem, for example, U.S. Pat. No. 11,374,467 discloses a power tool having a cyclone separation unit for collecting dust.

SUMMARY

[0004] The housing of the power tool disclosed in the U.S. Pat. No. 11,374,467 is provided with intake grilles corresponding to air inlets of the cyclone separation unit. Outside air entering the housing is led into the cyclone separation unit through the air inlets after large particles are filtered out through the intake grilles. This power tool leaves room for further improvement in terms of more effective dust collection.

[0005] It is accordingly a non-limiting object of the present disclosure to provide improved dust collecting performance in a power tool having a cyclone separation unit.

[0006] According to a non-limiting aspect of the present disclosure, a power tool is provided that is configured to drive a tool accessory removably mounted thereto. The power tool has a housing, a motor, a fan and a cyclone separation unit.

[0007] The housing has first air inlets. The motor is housed in the housing. The fan is housed in the housing. The fan is configured to be rotated by the motor and generate an air flow flowing in from the first air inlets and passing through the housing. The cyclone separation unit is housed in the housing. The cyclone separation unit has second air inlets arranged in a circumferential direction of the cyclone separation unit. The first air inlets of the housing and the second air inlets of the cyclone separation unit are arranged apart from each other in an axial direction of the cyclone separation unit. A first dust collecting chamber is defined: (i) between the first air inlets and the second air inlets in the axial direction of the cyclone separation unit, and (ii) between an inner peripheral surface of the housing and the cyclone separation unit, and configured to collect dust by inertia force. A second dust collecting chamber is defined within the cyclone separation unit and configured to collect dust by centrifugal force.

[0008] The power tool according to this aspect has two dust collecting chambers (the first dust collecting chamber and the second dust collecting chamber that is defined downstream of the first dust collecting chamber within the cyclone separation unit) in a path of the air flow that is generated by the fan and flows through the housing. In the first dust collecting chamber, dust having a relatively large particle size is collected by utilizing inertia force. In the second dust collecting chamber, dust having smaller particle size that are not collected in the first dust

collecting chamber is collectable by utilizing centrifugal force. Thus, in the power tool according to this aspect, dust is effectively collected in two stages.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- [0009] FIG. 1 is a top view of a grinder according to a first embodiment of the present disclosure.
[0010] FIG. 2 is a bottom view of the grinder.
[0011] FIG. 3 is a left side view of the grinder.
[0012] FIG. 4 is a right side view of the grinder.
[0013] FIG. 5 is a back view of the grinder.
[0014] FIG. 6 is a sectional view taken along line VI-VI in FIG. 1.
[0015] FIG. 7 is a sectional view of a cyclone separation unit.
[0016] FIG. 8 is a sectional view taken along line VIII-VIII in FIG. 7.
[0017] FIG. 9 is a perspective view of the cyclone separation unit.
[0018] FIG. 10 is an exploded perspective view of the cyclone separation unit.
[0019] FIG. 11 is a partial, enlarged view of FIG. 6, with a cover in a closed position and a locking member in a locking position.
[0020] FIG. 12 is a sectional view taken along line XII-XII in FIG. 5.
[0021] FIG. 13 is a partial, sectional view corresponding to FIG. 11, with the cover in an open position and the locking member in the locking position.
[0022] FIG. 14 is a partial, sectional view corresponding to FIG. 11, with the cover in the closed position and the locking member in an unlocking position.
[0023] FIG. 15 is a perspective view of a grinder according to a second embodiment, with a cover in the closed position.
[0024] FIG. 16 is a perspective view of the grinder, with the cover in the open position.
[0025] FIG. 17 is a partial, sectional view of the grinder, with the cover in the closed position.
[0026] FIG. 18 is a partial, sectional view of the grinder, with the cover in the open position.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0027] In one non-limiting embodiment according to the present disclosure, the housing may have at least one first outlet. The first outlet may be arranged between the first air inlets and the second air inlets in the axial direction of the cyclone separation unit and communicate the inside and outside of the first dust collecting chamber. According to this embodiment, dust collected in the first dust collecting chamber defined within the housing is dischargeable through the at least one first outlet.

[0028] In addition or in the alternative to the preceding embodiment, the at least one first outlet may be arranged so as not to overlap the second air inlets. According to this embodiment, air outside of the housing is prevented from directly flowing into the second air inlets of the cyclone separation unit without passing through the first dust collecting chamber.

[0029] In addition or in the alternative to the preceding embodiments, a normal attitude of the power tool in use may be defined relative to a direction of gravity. The at least one first outlet may be formed to face in the direction of gravity when the power tool is in the normal attitude. According to this embodiment, when the power tool is in the normal attitude, dust is dischargeable through the first dust outlet by utilizing the own weight of the dust.

[0030] In addition or in the alternative to the preceding embodiments, the power tool may be a grinder. The power tool may further have a spindle configured to be rotationally driven by the motor around a drive axis that defines an up-down direction of the power tool. The spindle may have a lower end part configured to removably hold the tool accessory. The housing may extend in a front-rear direction orthogonal to the up-down direction. The housing may be configured such

that an auxiliary handle is removably mounted on right and left sides of the housing in a left-right direction orthogonal to the up-down direction and the front-rear direction. The at least one first outlet may include at least one of (i) a lower outlet opened downward in the housing, (ii) a left outlet opened to the left in the housing, and (iii) a right outlet opened to the right in the housing. [0031] According to this embodiment, when the grinder is in an attitude in which the lower end of the spindle faces in the direction of gravity, dust is dischargeable through the lower outlet by utilizing the own weight of the dust. When the grinder is in an attitude in which the left side of the housing faces in the direction of gravity, with the auxiliary handle mounted on the right side of the housing, dust is dischargeable through the left outlet by utilizing the own weight of the dust. Further, when the grinder is in an attitude in which the right side of the housing faces in the direction of gravity, with the auxiliary handle mounted on the left side of the housing, dust is dischargeable through the right outlet by utilizing the own weight of the dust.

[0032] In addition or in the alternative to the preceding embodiments, the power tool may be a grinder. The power tool may further have a spindle configured to be rotationally driven by the motor around a drive axis that defines an up-down direction of the power tool. The spindle may have a lower end part configured to removably hold the tool accessory. The housing may extend in a front-rear direction orthogonal to the up-down direction. The first air inlets may be formed on right and left sides of the housing in a left-right direction orthogonal to the up-down direction and the front-rear direction.

[0033] When not in use, the grinder is often placed on ground, floor or a work table in an attitude in which an upper or lower end of the spindle faces in the direction of gravity. According to this embodiment, when the grinder is placed in this attitude, the possibility that dust scattering around the grinder and dropping from above enters the housing from the first air inlets is reduced.

[0034] In addition or in the alternative to the preceding embodiments, the power tool may further have a seal. The seal may be configured to seal a clearance between the housing and the cyclone separation unit on an opposite side of the first air inlets to the second air inlets in the axial direction of the cyclone separation unit. According to this embodiment, dust-containing air is prevented from flowing forward from the first dust collecting chamber without passing through the cyclone separation unit within the housing.

[0035] In addition or in the alternative to the preceding embodiments, the axial direction of the cyclone separation unit may define a front-rear direction of the power tool. The first air inlets may be arranged rearward of the second air inlets. The cyclone separation unit may include: (i) an outer cylinder, (ii) an intermediate cylinder that is arranged radially inside of the outer cylinder and has a rear end located forward of a rear end of the outer cylinder, and (iii) an inner cylinder that is arranged radially inside of the intermediate cylinder and has a rear end located forward of the rear end of the intermediate cylinder. The first dust collecting chamber may be defined between the housing and the outer cylinder. A dust separation chamber may be defined between the intermediate cylinder and the inner cylinder and communicate with the first dust collecting chamber via the second air inlets. An exhaust passage may be defined within the inner cylinder and communicate the dust separation chamber and the outside of the cyclone separation unit. The second dust collecting chamber may be defined between the outer cylinder and the intermediate cylinder and communicate with the dust separation chamber. According to this embodiment, the cyclone separation unit is provided that is configured to realize effective dust collection.

[0036] In addition or in the alternative to the preceding embodiments, the cyclone separation unit may be configured such that a first swirl flow is generated within the dust separation chamber and a second swirl flow is generated within the second dust collecting chamber. According to this embodiment, dust is effectively separated from the air by centrifugal force in both of the dust separation chamber and the second dust collecting chamber.

[0037] In addition or in the alternative to the preceding embodiments, the second dust collecting chamber may include: (i) a rear dust collecting chamber that is defined behind the intermediate

cylinder within the outer cylinder and communicates with the dust separation chamber, and (ii) a front dust collecting chamber that is defined between the outer cylinder and the intermediate cylinder in front of the rear dust collecting chamber and communicates with the rear dust collecting chamber. According to this embodiment, dust-containing air flowing out rearward from the intermediate cylinder is swirled forward around the intermediate cylinder within the front dust collecting chamber, so that dust is effectively collected in the front dust collecting chamber.

[0038] In addition or in the alternative to the preceding embodiments, the front dust collecting chamber may be configured to have a sectional area at least partially decreasing forward.

According to this embodiment, the flow velocity of air in the front dust collecting chamber is increased, so that dust is more efficiently separated by centrifugal force.

[0039] In addition or in the alternative to the preceding embodiments, the outer cylinder, the intermediate cylinder and the inner cylinder may be separately formed and integrally connected to each other to form a single separation unit assembly. According to this embodiment, the cyclone separation unit is easily mountable to the housing.

[0040] In addition or in the alternative to the preceding embodiments, the cyclone separation unit may be disposed in the housing so as to be removable from the housing. According to this embodiment, a user can remove the cyclone separation unit from the housing to dispose of the dust in a desired place or to clean the cyclone separation unit and/or the inside of the housing.

[0041] In addition or in the alternative to the preceding embodiments, the air flow generated by the fan may be cooling air for cooling the motor. The motor may be arranged downstream of the cyclone separation unit in a flow direction of the air flow. According to this embodiment, the possibility of a failure or trouble of the motor caused by dust contained in the motor cooling air is reduced.

[0042] In addition or in the alternative to the preceding embodiments, the power tool may be a grinder. The power tool may further have a spindle configured to be rotationally driven by the motor around a drive axis that defines an up-down direction of the power tool. The spindle may have a lower end part configured to removably hold the tool accessory. A rotational axis of the motor may extend in a front-rear direction orthogonal to the drive axis. The motor may be arranged behind the spindle. The cyclone separation unit may be arranged behind the motor. According to this embodiment, the cyclone separation unit is rationally arranged in the grinder.

First Embodiment

[0043] An electric disc grinder **1A** (hereinafter simply referred to as a grinder **1A**) according to a first embodiment is now described with reference to FIGS. **1** to **14**. The grinder **1A** is a representative example of a power tool of the present disclosure. More specifically, the grinder **1A** is a representative example of a rotary tool that is configured to perform processing operation (such as grinding, polishing and cutting) by rotationally driving a disc-like tool accessory **91** (such as a grinding wheel, a rubber pad, a brush and a blade). The grinder **1A** is also referred to as an angle grinder.

[0044] First, the structure of the grinder **1A** is described in belief.

[0045] As shown in FIGS. **1** to **6**, the grinder **1A** has an elongate housing **10A** that forms an outer shell of the grinder **1A**. The housing **10A** is also referred to as a body housing.

[0046] As shown in FIG. **6**, the housing **10A** houses a motor **21** and a spindle **25**. The motor **21** is arranged such that a rotational axis **RX** of an output shaft **215** of the motor **21** extends along a longitudinal axis of the housing **10A**. The spindle **25** is arranged within one end part of the housing **10A** in the longitudinal direction. The spindle **25** is operably connected to the output shaft **215** of the motor **21** and configured to be rotationally driven around a drive axis **DX** by the rotational power of the motor **21**. A power cord **19** that is connected to an external AC power source is connected to the other end part of the housing **10A** in the longitudinal direction. The grinder **1A** may however be provided with a battery mounting part configured to removably receive a rechargeable battery, in place of the power cord **19**.

[0047] One end part of the spindle **25** in the axial direction protrudes to the outside from the housing **10A**. This end part of the spindle **25** is configured to removably hold the tool accessory **91** and referred to as a tool mounting part **251**. The spindle **25** rotationally drives the tool accessory **91** mounted to the tool mounting part **251** when the motor **21** is driven.

[0048] The drive axis DX of the spindle **25** extends in a direction crossing the rotational axis RX of the output shaft **215** (more specifically, in a direction orthogonal to the rotational axis RX). In the following description, for convenience sake, the extending direction of the rotational axis RX of the output shaft **215** (the longitudinal direction of the housing **10A**) is defined as a front-rear direction of the grinder **1A**. In the front-rear direction, the side on which the spindle **25** is arranged is defined as a front side of the grinder **1A**, and the opposite side (the side on which the power cord **19** is arranged) is defined as a rear side of the grinder **1A**. The extending direction of the drive axis DX is defined as an up-down direction of the grinder **1A**. In the up-down direction, the side on which the tool mounting part **251** is arranged is defined as a lower side of the grinder **1A**, and the opposite side is defined as an upper side of the grinder **1A**. A direction orthogonal to the front-rear direction and the up-down direction is defined as a left-right direction of the grinder **1A**.

[0049] The structure of the grinder **1A** is now described in detail.

[0050] First, the structure of the housing **10A** is described. As shown in FIGS. **1** to **6**, the housing **10A** includes a gear housing **11**, a motor housing **13A** and a handle housing **15**.

[0051] The gear housing **11** forms a front end part of the housing **10A**. Air outlets **103** are formed in an upper part of the gear housing **11** to discharge air to the outside of the housing **10A**. In this embodiment, the air outlets **103** are formed through the gear housing **11** in the front-rear direction.

[0052] The gear housing **11** is configured such that a side handle (also referred to as an auxiliary handle) **95** is mountable thereto. More specifically, the gear housing **11** has a handle mounting part **111** formed as a female thread part. The side handle **95** includes an elongate grip part **951** configured to be held by a user, and a male thread part (not shown) protruding from one end of the grip part **951**. The side handle **95** is mounted to the gear housing **11** by screwing the male thread part into the handle mounting part **111** (female thread part). The side handle **95** may however be mounted to the gear housing **11** by any other method.

[0053] In this embodiment, the handle mounting part **111** is formed on the left, right and upper sides of the gear housing **11**. A user can selectively mount the side handle **95**, for auxiliary use, to any one of the three handle mounting parts **111** according to the dominant hand or the working environment. The grip part **951** protrudes leftward, rightward or upward from the housing **10A** when the side handle **95** is mounted to the gear housing **11** (the housing **10A**).

[0054] The motor housing **13A** has a hollow cylindrical shape. In this embodiment, the motor housing **13A** is a single (seamless) cylindrical member. The motor housing **13A** is fixed to a rear end part of the gear housing **11** and extends rearward from the gear housing **11**.

[0055] The handle housing **15** has a hollow cylindrical shape and is fixed to a rear end part **133** of the motor housing **13A** and extends rearward from the motor housing **13A**. A front half of the handle housing **15** is configured as a housing part for housing a part of a cyclone separation unit **5A** (described below) and a controller **20**. A rear half of the handle housing **15** is configured as a grip part **155** to be held by a user. In this embodiment, the handle housing **15** is formed by a left shell **15L** and a right shell **15R** being fixedly connected together in the left-right direction (see FIG. **1**).

[0056] A front end part **151** of the handle housing **15** has a cylindrical shape having substantially the same diameter as the motor housing **13A**. An intermediate part **152** of the handle housing **15** extends rearward from the front end part **151** and has a cup-like shape having a diameter gradually decreasing rearward. The front end part **151** and the intermediate part **152** form the front half of the handle housing **15**. Air inlets **101** are formed in the intermediate part **152** to allow air to flow into the housing **10A**. More specifically, the air inlets **101** for communicating the inside and outside of the housing **10A** are respectively formed in right and left parts of the handle housing **15**.

[0057] The grip part **155** (the rear half of the handle housing **15**) has a cylindrical shape having a

smaller diameter than the front half of the handle housing **15**. The grip part **155** is a part to be held by a user during processing operation using the grinder **1A**, and is also referred to as a main grip (main handle).

[0058] Elements disposed within the housing **10A** are now described.

[0059] As shown in FIG. **6**, the housing **10A** houses the motor **21**, a fan **23**, the spindle **25**, the controller **20**, a main switch **158** and a cyclone separation unit **5A**.

[0060] The motor **21** is disposed within the motor housing **13A** (substantially in the middle of the housing **10A**). The motor **21** has a body **210** including a stator **211** and a rotor **212**, and the output shaft **215** that integrally rotates with the rotor **212**. In this embodiment, the motor **21** is a commutator motor, but the kind of the motor **21** is not particularly limited. For example, it may be a brushless DC motor.

[0061] The fan **23** is fixed to the output shaft **215** in front of the body **210** of the motor **21**. The fan **23** generates an air flow (motor cooling air) for cooling the motor **21** by integrally rotating with the output shaft **215**. Specifically, the air flow generated by the fan **23** flows into the housing **10A** from the air inlets **101**, passes through the cyclone separation unit **5A** and the motor **21** and then flows to the outside from the air outlets **103**.

[0062] The spindle **25** is supported within the gear housing **11** (a front end part of the housing **10A**) so as to be rotatable around the drive axis DX. A large bevel gear is fixed to an upper end part of the spindle **25**. A front end part of the output shaft **215** of the motor **21** protrudes into the gear housing **11**. A small bevel gear is fixed to a front end part of the output shaft **215**. The small bevel gear is engaged with the large bevel gear of the spindle **25**. The tool mounting part **251** of the lower end part of the spindle **25** protrudes downward from the housing **10A** as described above.

[0063] The controller **20** is housed in the intermediate part **152** of the handle housing **15**. The controller **20** is configured to control operation (such as driving of the motor **21**) of the grinder **1A**. The controller **20** is located in a position at least partially overlapping the air inlets **101** of the housing **10A** when viewed from the left or the right. Thus, the controller **20** is effectively cooled by air flowing into the housing **10A** from the air inlets **101**.

[0064] A main switch **158** is provided for starting the motor **21** and housed within the grip part **155**. A trigger (also referred to as a switch lever) **157** is disposed on the lower side of the grip part **155**. The trigger **157** is a manual operation member configured such that a user can depress while holding the grip part **155**. The main switch **158** is configured to be normally in an OFF state and to be turned on while the trigger **157** is depressed. The main switch **158** is connected to the controller **20**. The controller **20** is configured to drive the motor **21** while the main switch **158** is in an ON state.

[0065] The cyclone separation unit **5A** is provided to swirl dust-containing air (hereinafter simply referred to as air) and separate dust from the air by utilizing centrifugal force. In this embodiment, as described above, the air flow (cooling air for cooling the motor **21**) generated by the fan **23** cools the motor **21** while flowing in from the air inlets **101** and passing through the housing **10A**, and flows out from the air outlets **103**. The cyclone separation unit **5A** is arranged between the air inlets **101** and the motor **21** in the path of the air flow within the housing **10A**. Specifically, in this embodiment, the cyclone separation unit **5A** is arranged in a position where it is separatable dust from the cooling air for the motor **21** before the cooling air reaches the motor **21**.

[0066] More specifically, the cyclone separation unit **5A** is arranged between the air inlets **101** and the motor **21** in the front-rear direction within the housing **10A**. Furthermore specifically, a front end of the cyclone separation unit **5A** is located rearward of the body **210** of the motor **21**, and a rear end of the cyclone separation unit **5A** is located forward of the air inlets **101**. The controller **20** is located substantially in the same position as the air inlets **101** in the front-rear direction, so that the rear end of the cyclone separation unit **5A** is located forward of the controller **20**. The cyclone separation unit **5A** has air inlets **51** in the front end part and is arranged such that an axis SX of the cyclone separation unit **5A** extends in the front-rear direction (the longitudinal direction of the

housing **10A**), which will be described in detail below.

[0067] In this embodiment, the cyclone separation unit **5A** is arranged to extend over the rear end part **133** of the motor housing **13A** and the front end part **151** of the handle housing **15**. Most of the cyclone separation unit **5A** is located within the rear end part **133** of the motor housing **13A**. A part of the housing **10A** that houses the cyclone separation unit **5A** is hereinafter referred to as a separation unit housing part **16**.

[0068] The structure of the cyclone separation unit **5A** is now described in detail.

[0069] As shown in FIGS. **7** and **8**, the cyclone separation unit **5A** has (i) air inlets **51** that communicates the inside and outside of the cyclone separation unit **5A**, (ii) a dust separation chamber **52** that communicates with the air inlets **51**, (iii) an exhaust passage **53** that communicates with the dust separation chamber **52** and the inside of the motor housing **13A**, (iv) a dust collecting chamber **54** that communicates with the dust separation chamber **52**, and (v) a dust outlet **55** that communicates the dust collecting chamber **54** and the outside of the cyclone separation unit **5A**.

[0070] The air inlets **51** are openings for allowing air to flow into the cyclone separation unit **5A**. In this embodiment, the air inlets **51** are arranged at equal intervals in a circumferential direction around the axis **SX** of the cyclone separation unit **5A** in the front end part of the cyclone separation unit **5A**. Air flows into the housing **10A** from the air inlets **101** and passes through the housing **10A** and then enters the air inlets **51**, which will be described in detail below.

[0071] The dust separation chamber **52** is a space (chamber) defined within the cyclone separation unit **5A** and has a conical cylindrical shape having a diameter gradually decreasing rearward from the inlets **51**. A first swirl flow flowing rearward is generated within the dust separation chamber **52** and dust is separated from the air by centrifugal force.

[0072] The exhaust passage **53** is an air passage defined radially inside (on the inner peripheral side) of the dust separation chamber **52**, and a rear end (inlet) of the exhaust passage **53** is located within the dust separation chamber **52**. Air from which dust is removed to some extent in the dust separation chamber **52** enters the exhaust passage **53** from the rear end of the exhaust passage **53** and flows forward and then enters the motor housing **13A** from a front end (outlet) of the exhaust passage **53**.

[0073] The dust collecting chamber **54** is a space (chamber) defined behind the dust separation chamber **52** and radially outside (on the outer peripheral side) of the dust separation chamber **52** within the cyclone separation unit **5A**. A second swirl flow is generated within the dust collecting chamber **54** and dust is separated from the air by centrifugal force. The dust separated from the air is collected in the dust collecting chamber **54**.

[0074] The dust outlet **55** is an opening provided to discharge dust collected in the dust collecting chamber **54** to the outside of the cyclone separation unit **5A**. In this embodiment, the dust outlet **55** is formed at a lower end of the cyclone separation unit **5A**.

[0075] Component members of the cyclone separation unit **5A** that define the air inlets **51**, the dust separation chamber **52**, the exhaust passage **53**, the dust collecting chamber **54** and the dust outlet **55** are now described. As shown in FIGS. **7** to **10**, the cyclone separation unit **5A** of this embodiment is formed by a first member **61**, a second member **62** and a third member **63** being connected to each other.

[0076] The first member **61** includes an inner cylinder part **611**, a base part **613**, guide vanes **615** and a connection part **617**. The inner cylinder part **611** is a circular cylindrical part having an axis extending in the front-rear direction and having both ends open in the axial direction. A front end part of the inner cylinder part **611** has a diameter gradually increasing forward. The base part **613** is a disc-like (annular) part protruding radially outward from the front end of the inner cylinder part **611**. The guide vanes **615** are arranged substantially at equal intervals in the circumferential direction. Each of the guide vanes **615** protrudes rearward from a rear surface of the base part **613** and extends in a tangential direction of a circle around the axis of the inner cylinder part **611**. The inner cylinder part **611** extends rearward of a rear end of the guide vane **615**. The connection part

617 is an elongate tubular part and has a threaded inner peripheral surface. The connection part **617** is supported by ribs protruding radially inward from an inner peripheral surface of the inner cylinder part **611** and extends rearward of the inner cylinder part **611** along the axis of the inner cylinder part **611**.

[0077] The second member **62** includes an intermediate cylinder part **621** and a flange part **623**. The intermediate cylinder part **621** is a conical cylinder part having an axis extending in the front-rear direction and having both ends open in the axial direction, and is arranged around (radially outside of) the inner cylinder part **611** of the first member **61**. The intermediate cylinder part **621** has an inner diameter larger than the outer diameter of the inner cylinder part **611** and gradually decreasing rearward. The flange part **623** protrudes radially outward from the front end of the intermediate cylinder part **621**. The second member **62** is arranged such that a front surface of the flange part **623** abuts on a receiving surface of the rear end of the guide vane **615** of the first member **61**. A rear end part of the inner cylinder part **611** of the first member **61** is arranged within the intermediate cylinder part **621**, and the rear end of the inner cylinder part **611** is located forward of the rear end of the intermediate cylinder part **621**. The connection part **617** of the first member **61** extends rearward of the intermediate cylinder part **621**.

[0078] The third member **63** includes an outer cylinder part **631**. The outer cylinder part **631** is a cylinder part having an axis extending in the front-rear direction and having one end open and the other end closed in the axial direction. More specifically, the outer cylinder part **631** includes a cylindrical peripheral wall part **632** and a circular bottom wall part **635** that closes a rear end of the peripheral wall part **632**.

[0079] The peripheral wall part **632** of the outer cylinder part **631** is arranged around (radially outside of) the intermediate cylinder part **621** of the second member **62**. The inner diameter of the peripheral wall part **632** is larger than the outer diameter of the intermediate cylinder part **621** and smaller than the outer diameter of the flange part **623**. The third member **63** is arranged such that a front end of the peripheral wall part **632** abuts on a rear surface of the flange part **623**. The intermediate cylinder part **621** of the second member **62** is arranged within the outer cylinder part **631**, and the rear end of the intermediate cylinder part **621** is located forward of the rear end of the outer cylinder part **631**. The outer diameter of the peripheral wall part **632** is smaller than the inner diameter of the separation unit housing part **16** (the rear end part **133** of the motor housing **13A** and the front end part **151** of the handle housing **15**) (see FIG. **11**).

[0080] One part of the peripheral wall part **632** protrudes radially outward of the other part. This part is hereinafter referred to as a protruding part **633**. More specifically, the protruding part **633** is formed as a part protruding downward on a lower front end part of the peripheral wall part **632**. The other part of the peripheral wall part **632** has substantially uniform inner and outer diameters. A rear end of the protruding part **633** is located forward of the rear end of the intermediate cylinder part **621**. The protruding part **633** has an opening formed through the protruding part **633** in the up-down direction (or an opening communicating the inside and outside of the peripheral wall part **632**). This opening is the dust outlet **55** provided to discharge dust to the outside of the cyclone separation unit **5A**.

[0081] A cover **71** for closing the dust outlet **55** and a locking mechanism **75** for the cover **71** are mounted to the cyclone separation unit **5A** (specifically, the outer cylinder part **631** of the third member **63**) of this embodiment. The cover **71** and the locking mechanism **75** will be described in detail below.

[0082] A through hole for a screw **65** is formed in the center of the bottom wall part **635** of the outer cylinder part **631**. A cylindrical receiving part **636** is formed around the through hole on a front surface of the bottom wall part **635**. A rear end part of the connection part **617** of the first member **61** is fitted in the receiving part **636**. In this state, the screw **65** is screwed into the threaded hole of the connection part **617** through the through hole from behind the bottom wall part **635**, so that the first, second and third members **61**, **62**, **63** are coaxially connected and fixed to each other.

Thus, in this embodiment, the first member **61**, the second member **62** and the third member **63** (including the cover **71** and the locking mechanism **75**) are coaxially connected and fixed to each other to form a single separation unit assembly **50** having the axis SX.

[0083] Further, in this embodiment, a seal **59** is mounted to the separation unit assembly **50** and integrated with the separation unit assembly **50**. More specifically, the seal **59** is fitted from the front and held onto a front end part (the base part **613** of the first member **61**) of the separation unit assembly **50**. The seal **59** is formed of an elastic material (such as rubber and synthetic resin foam). The seal **59** covers an outer periphery of the base part **613**. The seal **59** protrudes forward of a front end of the base part **613**.

[0084] Arrangement of the cyclone separation unit **5A** is now described. As shown in FIG. **11**, the cyclone separation unit **5A** is positioned such that the dust outlet **55** faces downward of the grinder **1A**, and held by the housing **10A**. In this embodiment, in the front-rear direction, most of the front end part of the cyclone separation unit **5A** is disposed in the rear end part **133** of the motor housing **13A**, and the rear end part of the cyclone separation unit **5A** is disposed in the front end part **151** of the handle housing **15**. The cyclone separation unit **5A** is coaxially arranged with the motor **21**.

[0085] When assembling the cyclone separation unit **5A** to the housing **10A**, first, the separation unit assembly **50** having the seal **59** mounted thereon is positioned in the circumferential direction such that the dust outlet **55** faces downward, and fitted into the cylindrical motor housing **13A** from the rear. The cyclone separation unit **5A** is positioned in the front-rear direction in a position where the seal **59** abuts on a protrusion **141** formed on the inside of the motor housing **13A**. The seal **59** seals a clearance between the front end part of the cyclone separation unit **5A** and an inner peripheral surface of the housing **10A** (the motor housing **13A**).

[0086] Thereafter, as shown in FIGS. **5** and **12**, the left and right shells **15L**, **15R** of the handle housing **15** are arranged to hold the rear end part of the cyclone separation unit **5A** in between and respectively fixed to the rear end part **133** of the motor housing **13A** by screws **140**. Further, the left and right shells **15L**, **15R** are connected to each other in the left-right direction by screws (not shown). In this manner, the cyclone separation unit **5A** formed as the separation unit assembly **50** is easily mountable to the handle housing **15**.

[0087] In this embodiment, the cyclone separation unit **5A** is positioned and held in the circumferential direction by protrusions **145** formed on the inside of the housing **10A** (the motor housing **13A** and the handle housing **15**) being engaged with recesses **501** formed in outer peripheral parts of the cyclone separation unit **5A** and the seal **59**. The protrusions **145** of the housing **10A** has a structure provided to connect the motor housing **13A** and the handle housing **15** by the screws **140**. Further, the cyclone separation unit **5A** is positioned and held in the front-rear direction by being sandwiched in the front-rear direction between the protrusion **141** of the motor housing **13A** and the protrusions **145** formed on the inside of the handle housing **15**.

[0088] With the above-described structure, as shown in FIG. **7**, spaces between the guide vanes **615** that are arranged between the rear surface of the base part **613** of the first member **61** and the front surface of the flange part **623** of the second member **62** in the front-rear direction form the air inlets **51** for allowing air to flow into the cyclone separation unit **5A**.

[0089] The dust separation chamber **52** is defined in the inside (between the intermediate cylinder part **621** and the inner cylinder part **611**) of the conical cylindrical intermediate cylinder part **621** of the second member **62**. As described above, the guide vanes **615** are arranged at equal intervals in the circumferential direction and each of the guide vanes **615** extends in a tangential direction of a circle around the axis SX (see FIG. **8**). Thus, air flowing in from the air inlets **51** efficiently forms the first swirl flow flowing rearward while being gathered and swirling around the inner cylinder part **611** in the dust separation chamber **52**. The first swirl flow flows rearward along an inner periphery of the conical intermediate cylinder part **621**. During this time, some of dust contained in the first swirl flow is collected and separated from the air by colliding with the inner periphery of the intermediate cylinder part **621** by centrifugal force.

[0090] The exhaust passage **53** is defined in the inside of the inner cylinder part **611** of the first member **61**. Air from which dust is separated as described above flows into the exhaust passage **53** from the inlet of the exhaust passage **53** (the rear end opening of the inner cylinder part **611**). The air flowing out from the outlet of the exhaust passage **53** (the front end opening of the inner cylinder part **611**) flows into the motor housing **13A**. This air cools the motor **21** while passing around the motor **21** and between the stator **211** and the rotor **212**, and flows to the outside from the air outlets **103** of the gear housing **11**. The above-described seal **59** seals the clearance between the front end part of the cyclone separation unit **5A** and the inner peripheral surface of the housing **10A**, so that the inside of the cyclone separation unit **5A** and the inside space of the motor housing **13A** substantially communicate with each other only via the exhaust passage **53**.

[0091] The dust collecting chamber **54** is defined in the inside of the outer cylinder part **631** of the third member **61** (between the outer cylinder part **631** and the intermediate cylinder part **621**). The dust collecting chamber **54** includes a region (hereinafter referred to as a rear dust collecting chamber **541**) behind the intermediate cylinder part **621** (the dust separation chamber **52**) and a region (hereinafter referred to as a front dust collecting chamber **543**) in front of the rear dust collecting chamber **541** and radially outside (on the outer peripheral side) of the intermediate cylinder part **621**. The swirling air flow flowing into the rear dust collecting chamber **541** from the dust separation chamber **52** forms the second swirl flow. The second swirl flow flows forward while swirling around the intermediate cylinder part **621** in the front dust collecting chamber **543**, and swirls in a front end part of the front dust collecting chamber **543**. Dust contained in the second swirl flow is collected by colliding with the inner periphery of the outer cylinder part **631** (the peripheral wall part **632**) by centrifugal force.

[0092] With the above-described configuration of the intermediate cylinder part **621** and the outer cylinder part **631**, the front dust collecting chamber **543** is configured to have a sectional area gradually decreasing forward. With this configuration, the flow velocity of air in the front dust collecting chamber **543** is increased, so that dust is more efficiently separated by centrifugal force. Further, a front end of the front dust collecting chamber **543** is closed by the flange part **623** of the second member **62**, so that dust stays mainly in the front end part of the front dust collecting chamber **543**.

[0093] The protruding part **633** is formed on the lower front end part of the outer cylinder part **631** as described above, so that the front end part of the front dust collecting chamber **543** includes a region protruding downward of the rear dust collecting chamber **541**. Dust entering this region easily stays without being relatively entrained in the second swirl flow in this region. This region is hereinafter referred to as a storage chamber **545**.

[0094] The dust outlet **55** is an opening formed in the protruding part **633** of the outer cylinder part **631** as described above, and communicates the dust collecting chamber **54** (the storage chamber **545** of the front dust collecting chamber **543**) and the outside of the cyclone separation unit **5A**.

[0095] As shown in FIGS. **2** and **11**, an opening **105** is formed through a lower end part of the housing **10A** in the up-down direction. The opening **105** is located right below the dust outlet **55** and communicates with the dust outlet **55**. Thus, the dust outlet **55** and the opening **105** communicate the inside of the cyclone separation unit **5A** and the outside of the housing **10A**.

[0096] In this embodiment, the opening **105** is formed to extend over the rear end part **133** of the motor housing **13A** and the front end part **151** of the handle housing **15**. More specifically, the opening **105** is formed by connecting a recess (notch) recessed forward from the rear end of the motor housing **13A** and a recess (notch) recessed rearward from the front end of the front end part **151** of the handle housing **15**. With this structure, compared with a structure in which the opening **105** is formed as a single through hole in either the motor housing **13A** or the handle housing **15**, the rigidity of the motor housing **13A** and the handle housing **15** is satisfactorily maintained.

[0097] The structures of the cover **71** and the locking mechanism **75** are now described in detail.

[0098] The cover **71** can be displaced between a closed position for closing the dust outlet **55** as

shown in FIGS. 7 and 11 and an open position for opening the dust outlet 55 as shown in FIG. 13. More specifically, the cover 71 of this embodiment is supported by the outer cylinder part 631 so as to be rotatable around a pivot axis PX between the closed position and the open position. The pivot axis PX extends in the left-right direction behind the dust outlet 55. The opening 105 of the housing 10A is configured not to disturb the opening and closing operation of the cover 71. In this embodiment, the opening 105 is configured such that the whole cover 71 is arranged within the range of the opening 105 in both of the closed position and the open position (see FIGS. 2, 11 and 13), when viewed from below the grinder 1A.

[0099] As shown in FIGS. 7 to 10, the cover 71 includes a body 711 that is configured to completely cover the dust outlet 55 from below when the cover 71 is in the closed position, a connection part 713 provided on a rear end of the body 711, and a holding plate 72 mounted on a lower side of the body 711.

[0100] In this embodiment, the dust outlet 55 substantially has a rectangular shape, and the body 711 has a corresponding rectangular shape. The connection part 713 includes two cylindrical parts extending in the left-right direction along the pivot axis PX. Two cylindrical support parts 715 are provided on the protruding part 633 of the outer cylinder part 631. The support parts 715 are respectively arranged on left and right sides of the connection part 713 of the cover 71. A rod 717 is inserted through the support parts 715 and the connection part 713 along the pivot axis PX. The cover 71 is rotatably supported by the support parts 715 via the rod 717. The holding plate 72 covers most of the body 711 from below. The holding plate 72 has a slot 721 extending in the front-rear direction.

[0101] The cover 71 is biased toward the open position by a biasing spring 73. The biasing spring 73 of this embodiment is a torsion coil spring (specifically, a double torsion spring) and is disposed around the rod 717. The biasing spring 73 is locked to the outer cylinder part 631 and the cover 71 and configured to bias the cover 71 in a direction (downward direction) of turning the cover 71 counterclockwise when viewed from the left relative to the outer cylinder part 631.

[0102] A seal 77 is mounted surrounding the dust outlet 55 on a lower end of the protruding part 633 of the outer cylinder part 631. The seal 77 is formed of an elastic material (such as rubber and synthetic resin foam). The seal 77 seals a clearance between the cover 71 (the body 711) and the dust outlet 55 when the cover 71 is in the closed position, and thereby prevents leakage of dust from the clearance between the cover 71 and the dust outlet 55.

[0103] The locking mechanism 75 is configured to be switched between a locking state in which the locking mechanism 75 locks the cover 71 in the closed position and an unlocking state in which it cannot lock the cover 71, according to user's manual operation. More specifically, as shown in FIGS. 7 to 10, the locking mechanism 75 of this embodiment includes a locking member 751, a receiving part 757 that can be engaged with the locking member 751, and a biasing spring 759 that biases the locking member 751.

[0104] The locking member 751 is supported by the cover 71 so as to be movable between a locking position and an unlocking position according to user's manual operation. The locking position is a position, as shown in FIGS. 7 to 9, where the locking member 751 is engageable with the receiving part 757 when the cover 71 is in the closed position. The unlocking position is a position, as shown in FIG. 14, where the locking member 751 is not engageable with the receiving part 757 when the cover 71 is in the closed position.

[0105] More specifically, the locking member 751 is supported between the body 711 of the cover 71 and the holding plate 72 so as to be slidable in the front-rear direction. The locking member 751 is an elongate member extending in the front-rear direction, and a front end part of the locking member 751 is configured as an engagement part 752 that can be engaged with the receiving part 757. The locking member 751 has a tab 753 protruding downward from a lower end of a central part of the locking member 751 in the front-rear direction. The tab 753 is an operation part configured to be manually operated by a user. The tab 753 protrudes downward from the holding

plate **72** through the slot **721** of the holding plate **72** and is slidable in the front-rear direction within the slot **721**. The user can hold the tab **753** and move the locking member **751** in the front-rear direction relative to the cover **71**.

[0106] In this embodiment, the tab **753** protrudes downward of a lower end of the housing **10A** through the opening **105** of the housing **10A** (see FIG. **11**). Thus, the user can easily operate the tab **753** even when it is hard to visually recognize the tab **753**.

[0107] The receiving part **757** protrudes downward from the protruding part **633** of the outer cylinder part **631** in front of the dust outlet **55** and has an engagement hole **758**. The engagement hole **758** is formed in a position facing the locking member **751** from the front when the cover **71** is in the closed position, and configured to receive the engagement part **752** of the locking member **751**.

[0108] The biasing spring **759** biases the locking member **751** forward (in a direction away from the pivot axis PX) relative to the cover **71**. More specifically, a recess **754** is formed in an upper part of the locking member **751** and extends forward from a rear end of the locking member **751** and has a closed front end. The biasing spring **759** is disposed within the recess **754**. A projection **712** is provided on a lower surface of the body **711** of the cover **71** and protrudes downward. The projection **712** abuts on a rear end of the biasing spring **759** within the recess **754**.

[0109] With such a structure, as shown in FIG. **7**, when the cover **71** is in the closed position, the locking member **751** is biased forward by the biasing spring **759**, and the engagement part **752** is engaged with the engagement hole **758** of the receiving part **757** in the locking position. Thus, the locking mechanism **75** is in the locking state. As shown in FIG. **14**, when the user pushes the tab **753** rearward and moves the locking member **751** rearward to the unlocking position against the biasing force of the biasing spring **759**, the engagement part **752** is disengaged from the engagement hole **758**. Thus, the locking mechanism **75** is switched into the unlocking state. When the user releases the tab **753** in this state, as shown in FIG. **13**, the cover **71** is biased to be rotated by the biasing spring **73** and rotated to the open position. Further, the locking member **751** is biased forward by the biasing spring **759** and placed in the locking position.

[0110] In processing operation using the grinder **1A**, a user selectively mounts the side handle **95** to any one of the three handle mounting parts **111** (see FIG. **1**) and holds the grip part **155** of the handle housing **15** with one hand and the side handle **95** with the other hand. The grinder **1A** can be used in various attitudes, but it is often used in an attitude in which the drive axis DX extends in the vertical direction and a lower end of the spindle **25** faces in the direction of gravity (downward in the vertical direction). Therefore, this attitude of the grinder **1A** is defined as a normal attitude of the grinder **1A**.

[0111] As shown in FIG. **11**, the dust outlet **55** of the cyclone separation unit **5A** and the opening **105** of the housing **10A** are formed to face in the direction of gravity when the grinder **1A** is in the normal attitude. Therefore, when the grinder **1A** is in the normal attitude, by operating the locking member **751** to open the cover **71**, a user can easily discharge dust collected in the dust collecting chamber **54** to the outside of the housing **10A** by utilizing the own weight of the dust. Particularly, in this embodiment, the cover **71** is automatically rotated to the open position by the biasing force of the biasing spring **759**, so that the user can open the cover **71** only by an operation of releasing locking of the locking member **751**.

[0112] In order to return the cover **71** to the closed position, the user pushes the tab **753** rearward and moves the locking member **751** rearward to the unlocking position against the biasing force of the biasing spring **759**, and further rotates the cover **71** to the closed position against the biasing force of the biasing spring **73**. When the user releases the tab **753** in this state, the locking member **751** is biased forward to the locking position by the biasing spring **759**, and the engagement part **752** is inserted into the engagement hole **758** of the receiving part **757**. Thus, the locking mechanism **75** is switched into the locking state.

[0113] In addition to the cyclone separation unit **5A**, the grinder **1A** of this embodiment further has

a structure for separating dust from air by utilizing inertia force upstream of the cyclone separation unit 5A in the air flow direction. Thus, the grinder 1A is configured to separate dust from the cooling air for the motor 21 in two stages. More specifically, as shown in FIG. 11, in this embodiment, a dust collecting chamber 18 is defined within the housing 10A and configured such that dust having a relatively large particle size can be separated therein by utilizing inertia force. [0114] The dust collecting chamber 18 is now described. The dust collecting chamber 18 is space (chamber) that is defined between the air inlets 101 of the housing 10A and the air inlets 51 of the cyclone separation unit 5A and between the inner peripheral surface of the housing 10A and the outer peripheral surface of the cyclone separation unit 5A, in the front-rear direction in the inside space of the housing 10A. As described above, the air inlets 101 are located rearward of the rear end of the cyclone separation unit 5A. Thus, the dust collecting chamber 18 includes a region behind the cyclone separation unit 5A and a region radially outside (on the outer peripheral side) of the cyclone separation unit 5A.

[0115] As described above, the seal 59 is provided to seal the clearance between the front end part of the cyclone separation unit 5A and the housing 10A in front of the air inlets 51 of the cyclone separation unit 5A. Thus, the dust collecting chamber 18 is separated from the inside space of the motor housing 13A. This prevents air flowing into the dust collecting chamber 18 from the air inlets 101 from directly flowing into the motor housing 13A without passing through the cyclone separation unit 5A.

[0116] Dust contained in the air flowing into the dust collecting chamber 18 from the air inlets 101 is collected by colliding with obstacles, such as the rear end surface of the cyclone separation unit 5A (the bottom wall part 635 of the outer cylinder part 631) and projections formed on the inner surface of the housing 10A, while flowing forward toward the air inlets 51 of the cyclone separation unit 5A.

[0117] As shown in FIGS. 1 to 3 and 11, the housing 10A has a dust outlet 180 provided to discharge dust collected in the dust collecting chamber 18 to the outside of the housing 10A. The dust outlet 180 is an opening that communicates the dust collecting chamber 18 and the outside of the housing 10A. In this embodiment, the housing 10A has three dust outlets 180. One of the dust outlets 180 is formed in the lower end part of the housing 10A and is hereinafter also referred to as a lower outlet 181. The other two dust outlets 180 are respectively formed in right and left parts of the housing 10A. The two dust outlets 180 are hereinafter also referred to as a left outlet 182 and a right outlet 183, respectively.

[0118] More specifically, the lower outlet 181 (see FIGS. 2 and 11) is a part of the opening 105 of the housing 10A (a clearance between the protruding part 633 and the handle housing 15) that is formed between the protruding part 633 of the cyclone separation unit 5A and the housing 10A behind the protruding part 633. The lower outlet 181 is configured to face in the direction of gravity when the grinder 1A is in the normal attitude. Therefore, when the grinder 1A is in the normal attitude, a user can easily discharge dust from the dust collecting chamber 18 to the outside of the housing 10A by utilizing the own weight of the dust.

[0119] The left outlet 182 (see FIGS. 1 and 2) is a slot formed in a left part of the rear end part 133 of the motor housing 13A and extending in the circumferential direction of the motor housing 13A. The right outlet 183 (see FIGS. 1 and 3) is a slot formed in a right part of the rear end part 133 of the motor housing 13A and extending in the circumferential direction of the motor housing 13A. The left and right outlets 182, 183 are arranged rearward of the air inlets 51 of the cyclone separation unit 5A.

[0120] Generally, when processing a floor surface extending substantially in the horizontal direction, a user mounts the side handle 95 to the left or right handle mounting part 111 of the gear housing 11 and uses the grinder 1A in an attitude in which the side handle 95 extends in a direction away from the floor surface (substantially upward in the direction of gravity). When the grinder 1A is in an attitude in which the side handle 95 mounted on the left side extends in a direction away

from the floor surface, the right outlet **183** faces in the direction of gravity, so that dust in the dust collecting chamber **18** can be easily discharged to the outside of the housing **10A** by utilizing the own weight of the dust. Similarly, when the grinder **1A** is in an attitude in which the side handle **95** mounted on the right side extends in a direction away from the floor surface, the left outlet **182** faces in the direction of gravity, so that dust in the dust collecting chamber **18** is easily dischargeable to the outside of the housing **10A** by utilizing the own weight of the dust.

[0121] All of the dust outlets **180** of the housing **10A** are arranged so as not to overlap the air inlets **51** of the cyclone separation unit **5A**. In other words, lines orthogonal to the axis SX of the cyclone separation unit **5A** and passing through the dust outlets **180** do not intersect with the air inlets **101**. This prevents air outside of the housing **10A** from directly flowing into the air inlets **51** of the cyclone separation unit **5A** without passing through the dust collecting chamber **18** of the housing **10A**.

[0122] As described above, the grinder **1A** of this embodiment has the dust collecting chamber **18** and the dust collecting chamber **54**, in the path of the air flow that is generated by the fan **23** and flows through the inside of the housing **10A**. In the dust collecting chamber **18** defined within the housing **10A** upstream of the cyclone separation unit **5A**, dust having a relatively large particle size is collected by utilizing inertia force. In the dust collecting chamber **54** within the cyclone separation unit **5A**, however, dust having smaller particle size that cannot be collected in the dust collecting chamber **18** can be collected by utilizing centrifugal force. Particularly, the amount of dust flowing into the cyclone separation unit **5A** is reduced by collecting some amount of dust in the dust collecting chamber **18**, so that the dust separation rate within the cyclone separation unit **5A** is improved. Thus, in the grinder **1A**, dust is effectively collected in two stages, so that the possibility of a failure or trouble of the motor **21** caused by dust contained in air flowing into the motor housing **13A** is reduced.

[0123] When not in use, the grinder **1A** is often placed on ground, floor or a work table in an attitude in which an upper or lower end of the spindle **25** faces in the direction of gravity. In this embodiment, the air inlets **101** are formed on the sides of the housing **10A**. This reduces the possibility that dust scattering around the grinder **1A** and dropping from above enters the housing **10A** from the air inlets **101** when the grinder **1A** is placed in the above-described attitude.

[0124] Further, the grinder **1A** of this embodiment has the cover **71** for opening and closing the dust outlet **55** of the cyclone separation unit **5A**, and the locking mechanism **75** configured to lock the cover **71** in the closed position. The locking mechanism **75** is not placed into the unlocking state unless the tab **753** of the locking member **751** is manually operated by a user. This reduces the possibility that dust is discharged from the dust outlet **55** at a time and a place not intended by the user.

[0125] Further, if a slide type cover is used to open and close the dust outlet **55**, dust may stay in a part that slidably holds the cover, and cause a failure or trouble in the opening and closing operation. In this embodiment, however, the cover **71** is a rotary type cover, and does not easily cause such a failure or trouble in opening and closing the cover **71** compared with the slide type cover. Further, a space required for the operation of opening and closing the cover **71** can be reduced in the axial direction of the cyclone separation unit **5A**, compared with the slide type cover. Furthermore, the pivot axis PX of the cover **71** is arranged behind the dust outlet **55** as described above, so that when the cover **71** is opened to discharge dust, the dust that scatters toward the user is reduced, compared with a structure in which the pivot axis is arranged in front of the dust outlet **55**.

Second Embodiment

[0126] An electric disc grinder **1B** (hereinafter simply referred to as a grinder **1B**) according to a second embodiment is now described with reference to FIGS. **15** to **18**. The grinder **1B** is different from the grinder **1A** of the first embodiment only in that a cyclone separation unit **5B** is removable from a housing **10B**. Therefore, component elements substantially identical to those of the first

embodiment are given the same numerals and their description is appropriately omitted or simplified, and different component elements or structures from the first embodiment are mainly described.

[0127] An elongate housing **10B** forms an outer shell of the grinder **1B**. The housing **10B** includes the gear housing **11**, a motor housing **13B** and the handle housing **15**. Although not shown in detail, like the first embodiment, the housing **10B** houses the motor **21**, the fan **23**, the spindle **25**, the controller **20**, the main switch **158** and a cyclone separation unit **5B** (see FIG. **6**). The handle housing **15** has the air inlets **101**, and the gear housing **11** has the air outlets **103**. The arrangement relation of these elements in the housing **10B** is substantially the same as that of the first embodiment.

[0128] In this embodiment, unlike the first embodiment, the housing **10B** has an opening **161** and a cover **165** configured to close the opening **161**. The opening **161** is formed in an upper half of a rear end part of the motor housing **13B**. The cyclone separation unit **5B** is housed in a part of the motor housing **13B** corresponding to the opening **161**.

[0129] The cyclone separation unit **5B** of this embodiment is shorter in the axial direction (the front-rear direction) than the cyclone separation unit **5A** of the first embodiment, but in the other points, the cyclone separation unit **5B** has substantially the same structure as the cyclone separation unit **5A** (see FIG. **11**). Most of the cyclone separation unit **5B** is arranged within the rear end part of the motor housing **13B**, and only a rear end part of the cyclone separation unit **5B** is located within the handle housing **15**. The cyclone separation unit **5B** is held in a position where the cover **71** and the locking mechanism **75** are arranged within the opening **105** of the housing **10B**, where the seal **59** fitted onto a front end part of the cyclone separation unit **5B** abuts on the protrusion **141** formed on the inside of the motor housing **13B**, and where a protrusion **146** formed on the inside of the handle housing **15** abuts on the rear end of the outer cylinder part **631**.

[0130] The cover **165** has a circular arc section corresponding to the upper half of the motor housing **13B**. The cover **165** can be displaced between a closed position for closing the opening **161** (see FIGS. **15** and **17**) and an open position for opening the opening **161** (see FIGS. **16** and **18**). More specifically, the cover **165** is supported by a left part of the motor housing **13B** so as to be rotatable around a pivot axis extending in the front-rear direction. A locking piece **166** is formed on a free end (right end) of the cover **165**. The locking piece **166** is flexible and has a claw **167** on its tip end. Although not shown in detail, a groove is formed on the inside of a right part of the motor housing **13B** so as to be engaged with the claw **167** of the locking piece **166**. The cover **165** is locked in the closed position by engagement of the claw **167** and the groove. When the cover **165** is in the closed position, like the first embodiment, the dust collecting chamber **18** is defined between the inner peripheral surface of the housing **10B** (including the cover **165**) and the outer peripheral surface of the cyclone separation unit **5B** within the housing **10B** (see FIG. **17**).

[0131] The opening **161** and the cover **165** may only be configured such that the cyclone separation unit **5B** is removable and such that the dust collecting chamber **18** is defined within the housing **10B** when the cover **165** is closed, and their sizes, shapes and opening/closing manners are arbitrarily changeable.

[0132] In this embodiment, a user can mount and remove the cyclone separation unit **5B** to and from the housing **10B** with the opening **161** opened by placing the cover **165** in the open position (see FIGS. **16** and **18**). For example, in order to remove the cyclone separation unit **5B**, the user moves the cyclone separation unit **5B** upward until the cover **71** and the locking mechanism **75** are separated from the opening **105**, and subsequently, takes the cyclone separation unit **5B** out of the housing **10B** through the opening **161** while pressing the cyclone separation unit **5B** forward so as to compress the seal **59**. The cyclone separation unit **5B** is mounted in a reverse procedure.

[0133] As described above, in the grinder **1B** of this embodiment, the cyclone separation unit **5B** is removable from the housing **10B**. Thus, the user can remove the cyclone separation unit **5B** from the housing **10B** and carry only the cyclone separation unit **5B** to a desired place to dispose of the

dust or to clean the cyclone separation unit **5B**, so that the convenience is improved.

[0134] In this embodiment, only the lower outlet **181** is provided as the dust outlet **180** that communicates the dust collecting chamber **18** and the outside. The user however can easily discharge dust collected in the dust collecting chamber **18** from the opening **105** or the opening **161** of the housing **10B** by removing the cyclone separation unit **5B**. Like the first embodiment, however, the dust outlet **180** may also be formed in right and left parts of the housing **10B** and/or the cover **165**.

[0135] Correspondences between the features of the above-described embodiments and the features of the present disclosure or invention are as follows. However, the features of the above-described embodiments are merely exemplary and do not limit the features of the present disclosure or invention.

[0136] The grinders **1A**, **1B** are examples of the “power tool” and the “grinder”. The housings **10A**, **10B** are examples of the “housing”. The air inlet **101** is an example of the “first air inlet”. The motor **21** and the fan **23** are examples of the “motor” and the “fan”, respectively. The cyclone separation units **5A**, **5B** are examples of the “cyclone separation unit”. The air inlet **51** is an example of the “second air inlet”. The dust collecting chamber **18** of the cyclone separation units **5A**, **5B** is an example of the “first dust collecting chamber”. The dust collecting chamber **54** of the cyclone separation units **5A**, **5B** is an example of the “second dust collecting chamber”.

[0137] The dust outlet **180** of the housings **10A**, **10B** is an example of the “first outlet”. The lower outlet **181**, the left outlet **182** and the right outlet **183** are examples of the “lower outlet”, the “left outlet” and the “right outlet”, respectively. The spindle **25** is an example of the “spindle”. The side handle **95** is an example of the “auxiliary handle”. The seal **59** is an example of the “seal”. The outer cylinder part **631** is an example of the “outer cylinder”. The intermediate cylinder part **621** is an example of the “intermediate cylinder”. The inner cylinder part **611** is an example of the “inner cylinder”. The dust separation chamber **52** is an example of the “dust separation chamber”. The exhaust passage **53** is an example of the “exhaust passage”. The rear dust collecting chamber **541** and the front dust collecting chamber **543** are examples of the “rear dust collecting chamber” and the “front dust collecting chamber”, respectively. The separation unit assembly **50** is an example of the “separation unit assembly”.

[0138] The above-described embodiments are merely exemplary, and the power tool according to the present disclosure is not limited to the grinders **1A**, **1B** of the above-described embodiments. For example, the following modifications may be made. At least one of these modifications can be adopted in combination with at least one of the features of the grinders **1A**, **1B** of the above-described embodiments and the claimed invention.

[0139] The power tool according to the present disclosure may be embodied not only as the grinders **1A**, **1B** but as other kinds of power tools to be used for processing operation that generates dust. Non-limiting examples of such power tools include a concrete planer, a hammer drill and a metal-cutting circular saw.

[0140] The housing of the power tool according to this disclosure may only be configured to have the first air inlets and define the first dust collecting chamber, and the shapes, numbers and positions of the members forming the housing and of the first air inlets and/or the shape and position of the first dust collecting chamber can be appropriately changed. For example, the motor housing **13A** (**13B**) and the handle housing **15** of the above-described embodiments may be integrally formed with each other by connecting their left shells and right shells.

[0141] The cyclone separation unit according to this disclosure need not be formed as an integral assembly by connecting a plurality of members as described above. For example, part of the cyclone separation unit may be integrally formed with the housing. The structure of the cyclone separation unit can be appropriately changed, insofar as it can separate dust from air flowing in from the air inlets and collect the dust in the dust collecting chamber by utilizing centrifugal force. For example, although the cyclone separation units **5A**, **5B** of the above-described embodiments

include three cylinder parts, the inner cylinder part **611**, the intermediate cylinder part **621** and the outer cylinder part **631**, the cyclone separation unit of this disclosure may only include at least (i) an outer cylinder that communicates with the air inlets and defines a dust collecting chamber in which dust can be separated from the air and collected, and (ii) an inner cylinder that is arranged radially inside of the outer cylinder and defines an exhaust passage for discharging the air from which the dust is separated in the dust collecting chamber.

[0142] In the above-described embodiments, the cover **71** and the locking mechanism **75** are provided in the cyclone separation units **5A**, **5B** and form part of the separation unit assembly **50**. The cover **71** and at least part of the mechanism for locking the cover **71** in the closed position may however be provided in the housing **10A**. For example, the cover **71** may be rotatably or slidably supported by the housing **10A**. The biasing spring **73** that biases the cover **71** toward the open position may be a spring of a different kind from the torsion coil spring (such as a plate spring and a compression coil spring), and the cover **71** need not be biased toward the open position. Further, the cover **71** may have an engagement part (such as a hole, a recess and a protrusion), and the locking member **751** may be configured to be engaged with the engagement part of the cover **71** and supported by the housing **10A** so as to be movable between the locking position and the unlocking position. The locking member **751** may be not a slide type but a rotary type, and need not be biased toward the locking position. The cover **71** and the locking mechanism **75** may be omitted.

[0143] In view of the nature of the present disclosure and the above-described embodiments, the following aspects are provided. At least one of the following aspects can be adopted in combination with at least one of the features of the above-described embodiments, its modifications and the claimed invention. [0144] (Aspect A1) The intermediate cylinder has a conical shape having a diameter gradually decreasing rearward, and [0145] at least part of the outer cylinder has a circular cylindrical shape having a substantially uniform diameter. [0146] (Aspect A2) A part of a front end part of the outer cylinder in a circumferential direction forms a protruding part protruding radially outward, and [0147] the dust outlet is formed in the protruding part.

[0148] The protruding part **633** of the above-described embodiments is an example of the “protruding part” of this aspect. [0149] (Aspect A3) The outer cylinder, the intermediate cylinder and the inner cylinder are coaxially fixed to each other by a screw.

[0150] The screw **65** of the above-described embodiments is an example of the “screw” of this aspect. [0151] (Aspect A4) An engagement part is provided on the inside of the housing and configured to be engaged with the cyclone separation unit so as to restrict movement of the cyclone separation unit at least either in the axial direction or the circumferential direction.

[0152] The protrusions **141**, **145**, **146** of the above-described embodiments are examples of the “engagement part” of this aspect. [0153] (Aspect A5) The power tool further comprises a controller configured to control driving of the motor, and [0154] the controller is arranged adjacent to the first air inlets within the housing.

[0155] The controller **20** of the above-described embodiments is an example of the “controller” of this aspect. [0156] (Aspect A6) The power tool comprises: [0157] a cover that is displaceable between a closed position for closing the dust outlet and an open position for opening the dust outlet, [0158] a spring that is configured to bias the cover toward the open position, and [0159] a locking mechanism that is configured to lock the cover in the closed position.

[0160] The cover **71**, the biasing spring **73** and the locking mechanism **75** are examples of the “cover”, the “spring, and the “locking mechanism” of this aspect, respectively. [0161] (Aspect A7) The locking mechanism includes an operation part configured to be manually operated by a user, and is configured to be switched between a locking state in which the locking mechanism locks the cover in the closed position and an unlocking state in which the locking mechanism does not lock the cover, according to user's manual operation of the operation part.

[0162] The locking member **751** (the tab **753**) of the above-described embodiments is an example

of the “operation part” of this aspect. [0163] (Aspect A8) The cover, the spring and the locking mechanism are integrally connected to the cyclone separation unit to form a single separation unit assembly.

[0164] Further, the following aspects B1 to B19 are provided as one non-limiting object to provide an improvement in a dust discharging mechanism of the cyclone separation unit of the power tool. Each of the following aspects B1 to B19 can be adopted alone or in combination with the other one or more of them. Alternatively, at least one of aspects B1 to B19 can be adopted in combination with at least one of the features of the grinders 1A, 1B of the above-described embodiments, the above-described modifications, aspects A1 to A8 and the claimed invention. [0165] (Aspect B1) [0166] A power tool, which is configured to drive a tool accessory that is removably mounted thereto, comprising: [0167] a motor; [0168] a housing that houses the motor; [0169] a cyclone separation unit that is held by the housing and has (i) air inlets, (ii) a dust collecting chamber that communicates with the air inlets, and (iii) a dust outlet that communicates the inside and outside of the dust collecting chamber; [0170] a cover that can be displaced between a closed position for closing the dust outlet and an open position for opening the dust outlet; [0171] a spring that is configured to bias the cover toward the open position; and [0172] a locking mechanism that is configured to lock the cover in the closed position.

[0173] According to aspect B1, the power tool effectively separates dust contained in an air flow by the cyclone separation unit. Further, by provision of the cover for opening and closing the dust outlet, and the locking mechanism configured to lock the cover in the closed position, the power tool reduces the possibility that dust is discharged at a time and a place not intended by the user. Furthermore, with the structure in which the cover is biased toward the open position, the dust outlet is opened simply by releasing locking of the locking mechanism. Thus, the power tool having excellent convenience is provided. [0174] (Aspect B2) [0175] The power tool as defined in aspect B1, wherein the locking mechanism (i) includes an operation part configured to be manually operated by a user, and (ii) is configured to be switched between a locking state in which the locking mechanism locks the cover in the closed position and an unlocking state in which the locking mechanism cannot lock the cover, according to user's manual operation of the operation part.

[0176] In the power tool according to aspect B2, the cover locked in the closed position is unlocked only when the operation part is manually operated by the user, so that the power tool more reliably reduces the possibility that dust is discharged at a time and a place not intended by the user. [0177] (Aspect B3) [0178] The power tool as defined in aspect B1 or B2, further comprising a seal that is configured to seal a clearance between the cover and the dust outlet.

[0179] According to aspect B3, the power tool prevents leakage of dust from the clearance between the cover and the dust outlet. [0180] (Aspect B4) [0181] The power tool as defined in any one of aspects B1 to B3, wherein: [0182] a rotational axis of the motor defines a front-rear direction of the power tool, [0183] the housing includes: (i) a first housing part that houses at least the motor and extends in the front-rear direction, and (ii) a second housing part that is connected to a rear end part of the first housing part and extends rearward, the second housing part including a grip part configured to be held by a user, [0184] the cyclone separation unit is held by the first and second housing parts behind the motor, and [0185] an opening is formed by connecting a recess formed at a rear end of the first housing part and a recess formed at a front end of the second housing part, and configured to expose the dust outlet of the cyclone separation unit to the outside of the housing when the cover is in the open position.

[0186] According to aspect B4, the recess (or notch) formed at the rear end of the first housing part forms a part of the opening exposing the dust outlet of the cyclone separation unit, and the recess (or notch) formed at the front end of the second housing part forms the remaining part of the opening. With this structure, compared with a structure in which the opening is formed as a single through hole in the housing, reduction of rigidity of the housing is restrained. [0187] (Aspect B5)

[0188] The power tool as defined in any one of aspects B1 to B4, wherein the cover, the spring and the locking mechanism are integrally connected to the cyclone separation unit to form a single separation unit assembly.

[0189] According to aspect B5, the cyclone separation unit, the cover, the spring and the locking mechanism are integrally assembled to the housing, so that the assemblability is improved. [0190] (Aspect B6) [0191] The power tool as defined in aspect B5, wherein the separation unit assembly is disposed in the housing so as to be removable from the housing.

[0192] According to aspect B6, the user can easily integrally remove the cyclone separation unit, the cover, the spring and the locking mechanism from the housing to dispose of the dust elsewhere.

[0193] (Aspect B7) [0194] The power tool as defined in any one of aspects B1 to B6, wherein the cover is rotatable around a pivot axis between the open position and the closed position.

[0195] According to aspect B7, by provision of the cover, compared with a slide type cover, dust does not easily stay in a movable part. [0196] (Aspect B8) [0197] The power tool as defined in aspect B7, wherein: [0198] the housing extends in a front-rear direction orthogonal to the up-down direction, [0199] a rear end part of the housing includes a grip part configured to be held by a user, and [0200] the pivot axis of the cover is located behind the dust outlet.

[0201] According to aspect B8, when the cover is opened to discharge dust, the dust that scatters toward the user is reduced, compared with a structure in which the pivot axis is located in front of the dust outlet. [0202] (Aspect B9) [0203] The power tool as defined in any one of aspects B1 to B8, wherein: [0204] a normal attitude of the power tool during driving of the tool accessory is defined relative to a direction of gravity, and [0205] the dust outlet is formed to face in the direction of gravity when the power tool is in the normal attitude.

[0206] According to aspect B9, when the power tool is in the normal attitude, dust can be discharged through the dust outlet by utilizing the own weight of the dust. [0207] (Aspect B10) [0208] The power tool as defined in any one of aspect B1 to B9, wherein: [0209] the axial direction of the cyclone separation unit defines a front-rear direction of the power tool, [0210] the cyclone separation unit includes: (i) an outer cylinder, (ii) an intermediate cylinder that is arranged radially inside of the outer cylinder and has a rear end located forward of a rear end of the outer cylinder, and (iii) an inner cylinder that is arranged radially inside of the intermediate cylinder and has a rear end located forward of the rear end of the intermediate cylinder, [0211] a dust separation chamber is defined between the intermediate cylinder and the inner cylinder and communicates with the outside of the cyclone separation unit via the air inlets, [0212] an exhaust passage is defined within the inner cylinder and communicates the dust separation chamber and the outside of the cyclone separation unit, and [0213] the dust collecting chamber is defined between the outer cylinder and the intermediate cylinder and communicates with the dust separation chamber.

[0214] According to aspect B10, the cyclone separation unit is provided that is configured to realize effective dust collection. [0215] (Aspect B11) [0216] The power tool as defined in aspect B10, wherein: [0217] the dust collecting chamber includes: (i) a rear dust collecting chamber that is defined behind the intermediate cylinder within the outer cylinder and communicates with the dust separation chamber, and (ii) a front dust collecting chamber that is defined between the outer cylinder and the intermediate cylinder in front of the rear dust collecting chamber and communicates with the rear dust collecting chamber, and [0218] the dust outlet communicates the inside and outside of the front dust collecting chamber.

[0219] According to aspect B11, dust-containing air flowing out rearward from the intermediate cylinder is swirled around the intermediate cylinder within the front dust collecting chamber, and dust is thereby effectively collected in the front dust collecting chamber and the collected dust is dischargeable to the outside through the dust outlet. [0220] (Aspect B12) [0221] The power tool as defined in B2 or any one of aspect B3 to B11 directly or indirectly dependent on aspect B2, wherein: [0222] the power tool is a grinder, [0223] the power tool further comprises a spindle configured to be rotationally driven by the motor around a drive axis that defines an up-down

direction of the power tool, the spindle having a lower end part configured to removably hold the tool accessory, [0224] a rotational axis of the motor extends in a front-rear direction orthogonal to the drive axis, [0225] the motor is arranged behind the spindle, [0226] the cyclone separation unit is arranged behind the motor, [0227] the dust outlet faces downward, and [0228] the operation part of the locking mechanism is configured to be manually operated from below the power tool. [0229] According to aspect B12, when a lower end of the spindle faces in the direction of gravity, dust can be discharged through the dust outlet by utilizing the own weight of the dust. Further, the possibility that the operation part is moved by unexpected external force to unlock the cover is reduced. [0230] (Aspect B13) [0231] The locking mechanism includes a locking member, [0232] the locking member is configured to be moved between (i) a locking position in which the locking member is lockable the cover when the cover is in the closed position, and (ii) an unlocking position in which the locking member is unlockable the cover when the cover is in the closed position, according to user's manual operation of the operation part. [0233] The locking member **751** of the above-described embodiments is an example of the “locking member” of this aspect. [0234] (Aspect B14) The locking member is supported by the cover so as to be movable between the locking position and the unlocking position, [0235] the housing or the cyclone separation unit has an engagement part that is configured to, when the cover is in the closed position and the locking member is in the locking position, be engaged with the locking member. [0236] The receiving part **757** (the engagement hole **758**) of the above-described embodiments is an example of the “engagement part” of this aspect. [0237] (Aspect B15) The locking mechanism includes a spring configured to bias the locking member toward the locking position. [0238] The biasing spring **759** of the above-described embodiments is an example of the “spring” of this aspect. [0239] (Aspect B16) The locking member is linearly movable between the locking position and the unlocking position, and [0240] the engagement part comprises a hole or a recess that is engaged with a part of the locking member. [0241] (Aspect B17) The housing has an opening that is configured to expose the dust outlet of the cyclone separation unit to the outside of the housing when the cover is in the open position, and [0242] the cover and at least part of the locking mechanism are arranged within the opening. [0243] (Aspect B18) The operation part protrudes downward from the opening of the housing. [0244] (Aspect B19) A part of the dust collecting chamber in a circumferential direction forms a protruding part protruding radially outward, and [0245] the dust outlet is formed in the protruding part. [0246] The protruding part **633** of the above-described embodiments is an example of the “protruding part” of this aspect. [0247] Correspondences between the features of aspects B1 to B19 and the features of the present disclosure or invention are as follows. However, the features of the above-described embodiments are merely exemplary and do not limit the features of aspects B1 to B19. [0248] The grinders **1A**, **1B** are examples of the “power tool” and the “grinder”. The motor **21** is an example of the “motor”. The housings **10A**, **10B** are examples of the “housing”. The cyclone separation units **5A**, **5B** are examples of the “cyclone separation unit”. The air inlet **51**, the dust collecting chamber **54** and the dust outlet **55** are examples of the “air inlet”, the “dust collecting chamber” and the “dust outlet”, respectively. The cover **71** is an example of the “cover”. The biasing spring **73** is an example of the “spring”. The locking mechanism **75** is an example of the “locking mechanism”. [0249] The tab **753** of the locking member **751** is an example of the “operation part”. The seal **77** is an example of the “seal”. The motor housings **13A**, **13B** are examples of the “first housing part”. The handle housing **15** and the grip part **155** are examples of the “second housing part” and the “grip part”, respectively. The opening **105** is an example of the “opening”. The separation unit assembly **50** is an example of the “separation unit assembly”. The pivot axis PX is an example of

the “pivot axis”. The outer cylinder part **631** is an example of the “outer cylinder”. The intermediate cylinder part **621** is an example of the “intermediate cylinder”. The inner cylinder part **611** is an example of the “inner cylinder”. The dust separation chamber **52** is an example of the “dust separation chamber”. The exhaust passage **53** is an example of the “exhaust passage”. The rear dust collecting chamber **541** and the front dust collecting chamber **543** are examples of the “rear dust collecting chamber” and the “front dust collecting chamber”, respectively. The spindle **25** is an example of the “spindle”.

[0250] The above-described embodiments are merely exemplary, and the power tool according to aspects B1 to B19 is not limited to the grinders **1A**, **1B** of the above-described embodiments. For example, the following modifications may be made. At least one of these modifications can be adopted in combination with at least one of the features of the grinders **1A**, **1B** of the above-described embodiments and the claimed invention.

[0251] The power tool according to aspects B1 to B19 may be embodied not only as the grinders **1A**, **1B** but as other kinds of power tools to be used for processing operation that generates dust. Non-limiting examples of such power tools include a concrete planer, a hammer drill and a metal-cutting circular saw.

[0252] The structure of the housing of the power tool according to this disclosure is not particularly limited, and the members forming the housing and the manner of connecting the members can be appropriately changed. For example, the motor housing **13A** (**13B**) and the handle housing **15** of the above-described embodiments may be integrally formed with each other by connecting their left shells and right shells. Further, the shape, number and position of the air inlets of the housing can be appropriately changed. For example, the air inlets **101** of the housing **10A** may be formed radially outward of the air inlets **51** of the cyclone separation unit **5A** (**5B**). In this modification, the dust collecting chamber **18** in the housing **10A** may be omitted.

[0253] The cyclone separation unit according to this disclosure need not be formed as an integral assembly by connecting a plurality of members as described above. For example, part of the cyclone separation unit may be integrally formed with the housing. The structure of the cyclone separation unit can be appropriately changed, insofar as it can separate dust from air flowing in from the air inlets and collect the dust in the dust collecting chamber by utilizing centrifugal force. For example, although the cyclone separation units **5A**, **5B** of the above-described embodiments include three cylinder parts, the inner cylinder part **611**, the intermediate cylinder part **621** and the outer cylinder part **631**, the cyclone separation unit of this disclosure may only include at least (i) an outer cylinder that communicates with the air inlets and defines a dust collecting chamber in which dust can be separated from the air and collected, and (ii) an inner cylinder that is arranged radially inside of the outer cylinder and defines an exhaust passage for discharging the air from which the dust is separated in the dust collecting chamber.

[0254] In the above-described embodiments, the cover **71** and the locking mechanism **75** are provided in the cyclone separation units **5A**, **5B** and form part of the separation unit assembly **50**. At least part of the mechanism for locking the cover **71** and the locking mechanism **75** in the closed position may however be provided in the housing **10A**. For example, the cover **71** may be rotatably or slidably supported by the housing **10A**. The biasing spring **73** that biases the cover **71** toward the open position may be a spring of a different kind from the torsion coil spring (such as a plate spring and a compression coil spring), and the cover **71** need not be biased toward the open position. Further, the cover **71** may have an engagement part (such as a hole, a recess and a protrusion), and the locking member **751** may be configured to be engaged with the engagement part of the cover **71** and supported by the housing **10A** so as to be movable between the locking position and the unlocking position. The locking member **751** may be not a slide type but a rotary type, and need not be biased toward the locking position.

DESCRIPTION OF THE REFERENCE NUMERALS

[0255] **1A**, **1B**: disc grinder (grinder), **10A**, **10B**: housing, **101**: air inlet, **103**: air outlet. **105**:

opening, 11: gear housing, 111: handle mounting part, 13A, 13B: motor housing, 133: rear end part, 140: screw, 141: protrusion, 145: protrusion, 146: protrusion, 15: handle housing, 15L: left shell, 15R: right shell, 151: front end part, 152: intermediate part, 155: grip part, 157: trigger, 158: main switch, 16: separation unit housing part, 161: opening, 165: cover, 166: locking piece, 167: claw, 18: dust collecting chamber, 180: dust outlet, 181: lower outlet, 182: left outlet, 183: right outlet, 19: power cord, 20: controller, 21: motor, 210: body, 211: stator, 212: rotor, 215: output shaft, 251: tool mounting part, 23: fan, 25: spindle, 5A, 5B: cyclone separation unit, 50: separation unit assembly, 501: recess, 51: air inlet, 52: dust separation chamber, 53: exhaust passage, 54: dust collecting chamber, 541: rear dust collecting chamber, 543: front dust collecting chamber, 545: storage chamber, 55: dust outlet, 59: seal, 61: first member, 611: inner cylinder part, 613: base part, 615: guide vane, 617: connection part, 62: second member, 621: intermediate cylinder part, 623: flange part, 63: third member, 631: outer cylinder part, 632: peripheral wall part, 633: protruding part, 635: bottom wall part, 636: receiving part, 65: screw, 71: cover, 711: body, 712: projection, 713: connection part, 715: support part, 717: rod, 72: holding plate, 721: slot, 73: biasing spring, 75: locking mechanism, 751: locking member, 752: engagement part, 753: tab, 754: recess, 757: receiving part, 758: engagement hole, 759: biasing spring, 77: seal, 91: tool accessory, 95: side handle, 951: grip part

Claims

1. A power tool, which is configured to drive a tool accessory that is removably mounted thereto, comprising: a housing having first air inlets; a motor that is housed in the housing; a fan that is housed in the housing and configured to be rotated by the motor and generate an air flow that flows in from the first air inlets and passes through the inside of the housing; and a cyclone separation unit that is housed in the housing and has second air inlets arranged in a circumferential direction of the cyclone separation unit; wherein: the first and second air inlets are arranged apart from each other in an axial direction of the cyclone separation unit, a first dust collecting chamber is defined: (i) between the first air inlets and the second air inlets in the axial direction of the cyclone separation unit, and (ii) between an inner peripheral surface of the housing and the cyclone separation unit, and configured to collect dust by inertia force, and a second dust collecting chamber is defined within the cyclone separation unit and configured to collect dust by centrifugal force.
2. The power tool as defined in claim 1, wherein the housing has at least one first outlet that is arranged between the first air inlets and the second air inlets in the axial direction of the cyclone separation unit and communicates the inside and outside of the first dust collecting chamber.
3. The power tool as defined in claim 2, wherein the at least one first outlet is arranged so as not to overlap the second air inlets.
4. The power tool as defined in claim 2, wherein: a normal attitude of the power tool in use is defined relative to a direction of gravity, and the at least one first outlet is formed to face in the direction of gravity when the power tool is in the normal attitude.
5. The power tool as defined in claim 2, wherein: the power tool is a grinder, the power tool further comprises a spindle configured to be rotationally driven by the motor around a drive axis that defines an up-down direction of the power tool, the spindle having a lower end part configured to removably hold the tool accessory, the housing extends in a front-rear direction orthogonal to the up-down direction, the housing is configured such that an auxiliary handle is removably mounted on right and left sides of the housing in a left-right direction orthogonal to the up-down direction and the front-rear direction, and the at least one first outlet includes at least one of (i) a lower outlet opened downward in the housing, (ii) a left outlet opened to the left in the housing, and (iii) a right outlet opened to the right in the housing.
6. The power tool as defined in claim 1, wherein: the power tool is a grinder, the power tool further

comprises a spindle configured to be rotationally driven by the motor around a drive axis that defines an up-down direction of the power tool, the spindle having a lower end part configured to removably hold the tool accessory, the housing extends in a front-rear direction orthogonal to the up-down direction, and the first air inlets are formed on right and left sides of the housing in a left-right direction orthogonal to the up-down direction and the front-rear direction.

7. The power tool as defined in claim 1, further comprising a seal that is configured to seal a clearance between the housing and the cyclone separation unit on an opposite side of the first air inlets to the second air inlets in the axial direction of the cyclone separation unit.

8. The power tool as defined in claim 1, wherein: the axial direction of the cyclone separation unit defines a front-rear direction of the power tool, the first air inlets are arranged rearward of the second air inlets, the cyclone separation unit includes: (i) an outer cylinder, (ii) an intermediate cylinder that is arranged radially inside of the outer cylinder and has a rear end located forward of a rear end of the outer cylinder, and (iii) an inner cylinder that is arranged radially inside of the intermediate cylinder and has a rear end located forward of the rear end of the intermediate cylinder, the first dust collecting chamber is defined between the housing and the outer cylinder, a dust separation chamber is defined between the intermediate cylinder and the inner cylinder and communicates with the first dust collecting chamber via the second air inlets, and an exhaust passage is defined within the inner cylinder and communicates the dust separation chamber and the outside of the cyclone separation unit, and the second dust collecting chamber is defined between the outer cylinder and the intermediate cylinder and communicates with the dust separation chamber.

9. The power tool as defined in claim 8, wherein the cyclone separation unit is configured such that a first swirl flow is generated within the dust separation chamber and a second swirl flow is generated within the second dust collecting chamber.

10. The power tool as defined in claim 8, wherein the second dust collecting chamber includes: (i) a rear dust collecting chamber that is defined behind the intermediate cylinder within the outer cylinder and communicates with the dust separation chamber, and (ii) a front dust collecting chamber that is defined between the outer cylinder and the intermediate cylinder in front of the rear dust collecting chamber and communicates with the rear dust collecting chamber.

11. The power tool as defined in claim 10, wherein the front dust collecting chamber is configured to have a sectional area at least partially decreasing forward.

12. The power tool as defined in claim 8, wherein the outer cylinder, the intermediate cylinder and the inner cylinder are separately formed and integrally connected to each other to form a single separation unit assembly.

13. The power tool as defined in claim 1, wherein the cyclone separation unit is disposed in the housing so as to be removable from the housing.

14. The power tool as defined in claim 1, wherein: the air flow generated by the fan is cooling air for cooling the motor, and the motor is arranged downstream of the cyclone separation unit in a flow direction of the air flow.

15. The power tool as defined in claim 14, wherein: the power tool is a grinder, the power tool further comprises a spindle configured to be rotationally driven by the motor around a drive axis that defines an up-down direction of the power tool, the spindle having a lower end part configured to removably hold the tool accessory, a rotational axis of the motor extends in a front-rear direction orthogonal to the drive axis, the motor is arranged behind the spindle, and the cyclone separation unit is arranged behind the motor.

16. The power tool as defined in claim 8, wherein: the intermediate cylinder has a conical shape having a diameter gradually decreasing rearward, and at least part of the outer cylinder has a circular cylindrical shape having a substantially uniform diameter.
